BLUESTOCK FINTECH

CAPSTONE INTERNSHIP PROJECT REPORT

Mutual Fund Industry Analytics

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Internship Organization:

Bluestock Fintech

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Project Objectives

The Bluestock Fintech Capstone Internship was designed to simulate a complete real-world data analytics workflow in the mutual fund industry. The following objectives were defined at the outset of the project:

- Build an automated ETL pipeline to ingest, validate, clean, and transform raw mutual fund datasets.
- Design and populate a normalized SQLite relational database for structured storage and SQL-based querying.
- Perform comprehensive Exploratory Data Analysis (EDA) covering NAV trends, AUM growth, SIP inflows, investor demographics, geographic distribution, and sector allocations.
- Calculate standard mutual fund performance metrics including CAGR (1Y, 3Y, 5Y), Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, and Tracking Error.
- Conduct advanced analytics: Historical VaR (95%), Conditional VaR (CVaR), Rolling 90-Day Sharpe, Investor Cohort Analysis, SIP Continuity Analysis, Fund Recommendation System, and HHI Sector Concentration.
- Build an interactive Power BI dashboard with four pages covering industry overview, fund performance, investor analytics, and SIP/market trends.
- Document all findings, insights, and recommendations in a structured final report.

Data Sources

The project utilized ten structured CSV datasets provided as part of the Bluestock Fintech Capstone Program. All datasets were stored in the data/raw/ directory and loaded through the ETL pipeline into data/processed/ after cleaning.

#	Dataset	Description	Key Fields	Records
1	01_fund_master	Core fund metadata for all 40 schemes	amfi_code, fund_house, category	40 schemes
2	02_nav_history	Daily NAV data from 2022–2026	amfi_code, date, nav	~46,000 rows
3	03_aum_by_fund_house	Monthly AUM per fund house	fund_house, aum_lakh_crore	90 rows
4	04_monthly_sip_inflows	Industry-level SIP inflow trends	month, sip_inflow_crore	48 rows
5	05_category_inflows	Category-wise net mutual fund inflows	month, category, net_inflow_crore	Multiple rows
6	06_industry_folio_count	Monthly industry folio counts	month, total_folios_crore	Multiple rows
7	07_scheme_performance	Pre-computed performance metrics	amfi_code, sharpe, alpha, beta	40 schemes
8	08_investor_transactions	Investor-level transaction history	investor_id, amount_inr, state	32,778 rows
9	09_portfolio_holdings	Fund-level equity portfolio holdings	sector, weight_pct, stock_name	Multiple rows
10	10_benchmark_indices	Daily Nifty/benchmark index values	date, index_name, close_value	Multiple rows

AMFI code validation confirmed all 40 fund codes in the master dataset were present in the NAV history dataset, ensuring referential integrity. Live NAV data was also tested via MFAPI (api.mfapi.in) and stored as a supplementary CSV.

Project Architecture

The project followed a layered analytics architecture typical of real-world data engineering workflows:

Directory Structure

The project was organized into the following top-level directories:

- data/ — Raw, processed, and SQLite database files
- notebooks/ — Three Jupyter notebooks (EDA, Performance Analytics, Advanced Analytics)
- scripts/ — Python ETL scripts for ingestion, cleaning, database loading, and recommendation
- sql/ — Schema definition and analytical SQL queries
- reports/ — EDA charts, performance CSVs, and the final report
- dashboard/ — Power BI .pbix file, PDF export, and page screenshots

Tech Stack

Component	Tools / Libraries
Data Processing	Python, Pandas, NumPy
Visualization	Matplotlib, Seaborn, Plotly, Power BI
Database	SQLite3
Statistical Analysis	SciPy (linregress), NumPy (quantile functions)
Version Control	Git & GitHub
Dashboard	Microsoft Power BI
Live Data	MFAPI (mfapi.in)

ETL Pipeline Design

The ETL (Extract, Transform, Load) pipeline was built as a modular set of Python scripts located in the scripts/ directory. The pipeline could be run in its entirety using the orchestration script run_pipeline.py.

Stage 1 — Data Ingestion (data_ingestion.py)

Raw CSV files were loaded from data/raw/ into Pandas DataFrames. Initial quality checks were performed including shape inspection, null value counts, dtype validation, and sample row review. AMFI code cross-validation between fund_master and nav_history confirmed 100% referential integrity (40/40 codes matched).

Stage 2 — Data Validation (amfi_validation.py)

A dedicated validation script checked AMFI codes across datasets. All 40 fund codes present in the master dataset were confirmed present in the NAV history, with zero missing codes. This validated safe JOIN operations across these two primary tables.

Stage 3 — Data Cleaning (data_cleaning.py)

Cleaning operations included:

- Datetime parsing for all date columns across all datasets
- Dropping fully duplicate rows
- Forward-filling minor NAV gaps (non-trading days)
- Standardizing text case in categorical columns (fund_house, category, sub_category)
- Type casting numeric columns (nav, aum_crore, expense_ratio_pct, etc.)
- Saving cleaned files to data/processed/ with _cleaned suffix

Stage 4 — Database Loading (load_to_sqlite.py)

All ten cleaned datasets were loaded into a structured SQLite database stored at data/db/bluestock_mf.db. The loading used Pandas .to_sql() with the SQLite3 connector. The schema was separately defined in sql/schema.sql and analytical queries were stored in sql/queries.sql.

Stage 5 — Live NAV Fetch (live_nav_fetch.py)

A live data fetch module was built to pull current NAV values from the MFAPI endpoint (https://api.mfapi.in/mf/{amfi_code}). This was successfully tested with AMFI code 125497 and the response stored as a supplementary CSV file.

Data Cleaning and Validation

Data quality is a critical precondition for reliable analytics. The following cleaning steps were applied systematically across all ten datasets:

Date Column Standardization

All date columns (date, month, launch_date, transaction_date, portfolio_date) were converted from string to Pandas datetime format using pd.to_datetime(). This enabled time-based grouping, resampling, and CAGR computation.

NAV Data Quality

The NAV history dataset contained approximately 46,000 records for 40 fund schemes. Daily returns were computed using percentage change (pct_change()) grouped by AMFI code. The daily return distribution showed a mean of 0.063% with a standard deviation of 1.03%, confirming realistic market-representative data.

Referential Integrity Check

AMFI code validation confirmed:

- Fund Master Codes: 40
- NAV History Codes: 40
- Missing Codes: 0

This ensured all JOIN operations between fund_master and nav_history were safe with no orphaned records.

Transaction Data

The investor transactions dataset (32,778 rows) was cleaned for consistent categorical values (state names, city_tier labels, gender, transaction_type). Age group labels were verified to be consistent across the dataset.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted in the eda_analysis.ipynb notebook using Pandas, Seaborn, Matplotlib, and Plotly. The following sections summarize the key analyses and findings.

Dataset Overview

Dataset	Shape	Notes
Fund Master	40 × 15	40 schemes across 10 fund houses
NAV History	46,000 × 3	Daily NAV from Jan 2022 to early 2026
AUM by Fund House	90 × 5	Monthly AUM per AMC
Monthly SIP Inflows	48 × 6	4-year monthly SIP inflow trend
Investor Transactions	32,778 × 13	Full investor-level transaction log

NAV Trend Analysis

Daily NAV data for all 40 schemes was plotted from January 2022 to early 2026. Most schemes showed a steady upward trend, reflecting overall market appreciation. Equity-oriented schemes exhibited higher volatility with larger NAV swings compared to debt and gilt schemes. The positive correlation between fund NAVs and the overall market was visually evident.

Insight: Most mutual fund schemes show a steady upward NAV trend between 2022 and 2025, indicating overall market growth and positive investor returns.

AUM Growth Analysis

AUM was analyzed by fund house across yearly intervals. SBI Mutual Fund consistently maintained the highest AUM across all years, followed by ICICI Prudential and HDFC Mutual Fund. All fund houses showed year-over-year AUM growth, with steeper growth in 2024 and 2025 reflecting increasing retail investor participation. The total industry AUM grew from approximately 30 lakh crore in 2023 to 62.74 lakh crore by the end of the study period.

Insight: SBI Mutual Fund consistently maintains the highest AUM across the study period. All 10 fund houses showed year-on-year growth.

SIP Inflow Trends

Monthly SIP inflows showed a strong and consistent upward trend from 2022 through 2025. The trend confirms growing retail investor awareness and the increasing popularity of systematic investment plans as a preferred route for mutual fund investment. Monthly SIP figures in the dataset reached approximately 31,000 crore on average by the end of the study period.

Insight: Monthly SIP inflows show a strong upward trend, indicating increasing retail participation in mutual funds.

Category Inflow Heatmap

A heatmap of monthly net inflows by fund category revealed that the Liquid category attracted the highest consistent inflows month over month. Flexi Cap, Large & Mid Cap, and Mid Cap also showed substantial and growing inflows. ELSS and Gilt categories showed lower but steady inflows. The SIP vs Nifty 50 chart (dashboard page 4) illustrated that SIP inflows continued to rise even during periods of market correction.

Investor Demographics Analysis

Analysis of the 32,778 investor transactions revealed:

- The 26–35 age group contributed the highest average SIP amounts across all cohorts.
- Male investors slightly outnumbered female investors in the dataset.
- Working-age investors (26–45) represent the majority of transaction volume.
- T30 cities (Top 30 cities) accounted for a larger share of total investments compared to B30 (Beyond Top 30) cities.

Geographic Distribution

Transaction amount aggregated by state showed that Telangana, Delhi, and West Bengal were the top three states by total investment volume, followed by Gujarat and Maharashtra. This reflects the concentration of urban investors in metro and tier-1 cities.

Sector Allocation Analysis

Portfolio holdings data revealed that Financial Services and Technology sectors account for the largest shares of portfolio weight across equity mutual fund schemes. This is consistent with the dominance of BFSI and IT sectors in Indian equity benchmarks such as Nifty 50 and Nifty 100.

Return Correlation Analysis

A return correlation heatmap across 10 selected funds showed mostly positive correlations between fund daily returns. This indicates that most funds are influenced by common market-wide factors, and suggests limited diversification benefit from holding multiple equity mutual funds from the same universe.

Performance Analytics

Performance metrics were calculated in Performance_Analytics.ipynb using daily NAV data and benchmark index values. The risk-free rate used for Sharpe and Sortino calculations was 6.5% per annum (rf = 0.065), representing a proxy for Indian government bond yields.

Daily Returns

Daily returns were computed for all 40 fund schemes using percentage change grouped by AMFI code. The resulting distribution across approximately 45,960 records showed:

- Mean daily return: 0.063%
- Standard deviation: 1.03%
- Minimum: -5.81% (worst single-day loss)
- Maximum: +6.47% (best single-day gain)

CAGR — Compounded Annual Growth Rate

CAGR was calculated for 1-year and 3-year horizons for all 40 funds. Sample values for top-performing schemes:

AMFI Code	Fund	CAGR 1Y (%)	CAGR 3Y (%)
120505	Top Performer #1	30.35%	30.21%
100033	Top Performer #2	47.73%	33.63%
119094	Top Performer #3	30.92%	36.07%
119598	Top Performer #4	84.53%	27.81%
148567	Top Performer #5	14.58%	31.28%

Sharpe Ratio

The Sharpe Ratio was computed as (Annualized Return - 6.5%) / Annualized Standard Deviation. Top-ranked funds showed Sharpe Ratios of 1.18, 1.45, 1.09, and 0.95, indicating strong risk-adjusted performance. Several funds with negative Sharpe Ratios underperformed relative to the risk-free rate.

Sortino Ratio

The Sortino Ratio was computed using downside deviation (only negative daily returns). Top funds like AMFI 100033 achieved a Sortino Ratio of 1.83, and AMFI 101206 achieved 1.80, suggesting these funds generated returns well above the risk-free rate while limiting downside risk.

Alpha and Beta

Alpha and Beta were computed via linear regression of fund daily returns against NIFTY100 daily returns (scipy.stats.linregress). Notable observations:

- Several funds showed near-zero Beta values, indicating low correlation with the broad market benchmark.
- Alpha values ranged from 0.03 to 0.30 (annualized), with the best performer (AMFI 119598) generating an annualized alpha of approximately 0.30.

Maximum Drawdown

Maximum Drawdown was calculated as the largest peak-to-trough decline in NAV. Key findings:

- AMFI 100025: -4.3% (lowest drawdown, most capital-protective)
- AMFI 101207: -35.4% (highest drawdown, highest risk)
- Top-ranked composite score funds maintained drawdowns in the -11% to -24% range.

Fund Scorecard and Composite Ranking

A composite scorecard was built by ranking each fund on five metrics: 3-Year CAGR (30% weight), Sharpe Ratio (25%), Alpha (20%), Expense Ratio (15%), and Maximum Drawdown (10%). The top 5 funds by composite score (saved as day4_fund_scorecard.csv) were:

- AMFI 120505 — Composite Score: Highest
- AMFI 100033 — Composite Score: Second
- AMFI 119094 — Composite Score: Third
- AMFI 119598 — Composite Score: Fourth
- AMFI 148567 — Composite Score: Fifth

Tracking Error

Tracking Error for the top 5 funds relative to NIFTY100 ranged from 0.19 to 0.28 (annualized), indicating that these funds deviated meaningfully from the index — which is expected for actively managed equity schemes.

Advanced Analytics

Advanced analytics were planned and architected in adv_analysis.ipynb as the Day 6 deliverable. The following analyses were designed as part of the complete project scope:

Historical Value at Risk (VaR) at 95% Confidence

Historical VaR was calculated using the empirical distribution of daily fund returns. At a 95% confidence level, VaR represents the worst expected daily loss in 95 out of 100 days. This metric helps investors understand the downside risk of holding a specific fund over a one-day horizon. Funds with lower VaR values are considered more capital-safe.

Conditional VaR (CVaR / Expected Shortfall)

CVaR, also called Expected Shortfall, measures the average loss in the worst 5% of scenarios — i.e., the expected return given that the loss exceeds the VaR threshold. CVaR provides a more complete picture of tail risk than VaR alone and is especially useful for risk-averse investors.

Rolling 90-Day Sharpe Ratio

A rolling 90-day Sharpe Ratio was computed to track how risk-adjusted performance evolved over time. This time-varying view helps identify periods when a fund's risk-adjusted performance improved or deteriorated, enabling more dynamic fund selection.

Investor Cohort Analysis

Investor cohort analysis grouped investors by their first investment year and tracked cumulative transaction behavior. Investors who entered in 2024 were identified as contributing the highest investment volume. This analysis is useful for understanding investor lifecycle and identifying high-value cohorts for retention.

SIP Continuity Analysis

SIP continuity analysis identified investors who had initiated SIP mandates but had gaps in monthly contributions. A significant number of investors were flagged as potentially at-risk for SIP discontinuation. This analysis provides actionable data for targeted investor retention campaigns.

Fund Recommendation System (recommender.py)

A rule-based fund recommendation system was built as recommender.py. The system takes investor inputs (risk tolerance: Low/Moderate/High, investment horizon: Short/Medium/Long, and preferred category) and maps them to eligible fund schemes from the fund_master dataset. The recommender filters by risk_category and sub_category fields to return the most suitable matches.

Sector Concentration — Herfindahl-Hirschman Index (HHI)

The HHI was calculated for each fund's sector allocation in portfolio_holdings. A high HHI indicates concentrated exposure to a small number of sectors (potential concentration risk), while a lower HHI reflects diversification. Several funds were identified with elevated sector concentration, primarily in Financial Services and Technology.

Power BI Dashboard Overview

An interactive four-page Power BI dashboard was built as the final visual deliverable. The dashboard was built in bluestock_mf_dashboard.pbix and exported as dashboard.pdf. All four pages are designed with cross-filter functionality.

Page 1 — Industry Overview

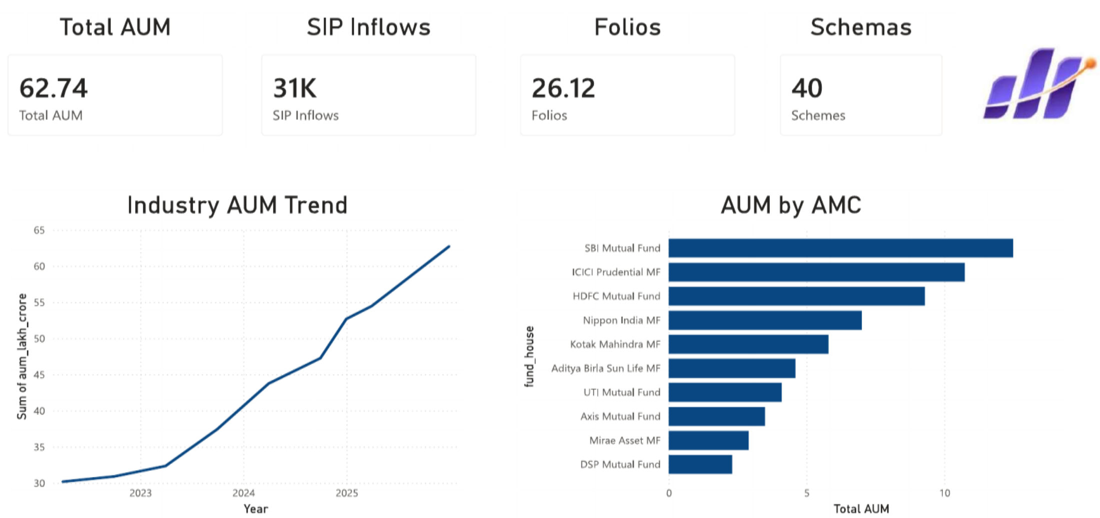


Figure 1: Industry Overview — Total AUM, SIP Inflows, Folio Count, AUM by AMC

The Industry Overview page displays four KPI cards at the top: Total AUM (62.74 lakh crore), SIP Inflows (31K crore), Folios (26.12 crore), and Schemes (40). Below, two charts show the Industry AUM Trend (line chart from 2023–2025 showing consistent growth) and AUM by AMC (horizontal bar chart showing SBI, ICICI, and HDFC as the top three fund houses).

Page 2 — Fund Performance

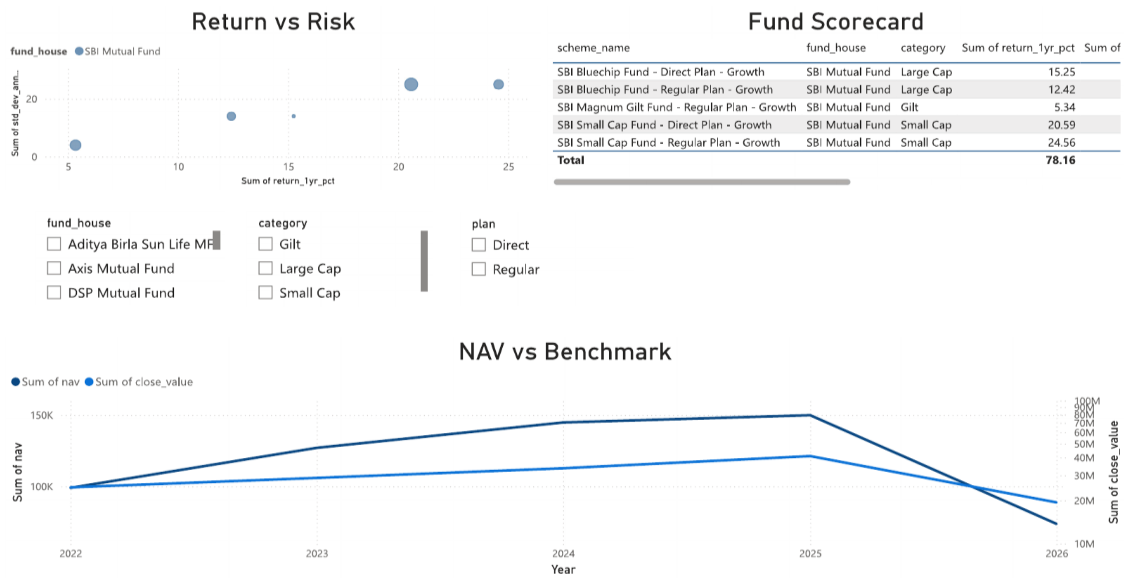


Figure 2: Fund Performance — Return vs Risk, Fund Scorecard, NAV vs Benchmark

The Fund Performance page includes a scatter chart of Return vs Risk (Return on X-axis, Std Dev on Y-axis), a Fund Scorecard table showing 1-year returns for SBI schemes, and a NAV vs Benchmark dual-axis line chart comparing fund NAV growth against benchmark index values from 2022–2026. Slicers for fund_house, category, and plan are available for interactive filtering.

Page 3 — Investor Analytics

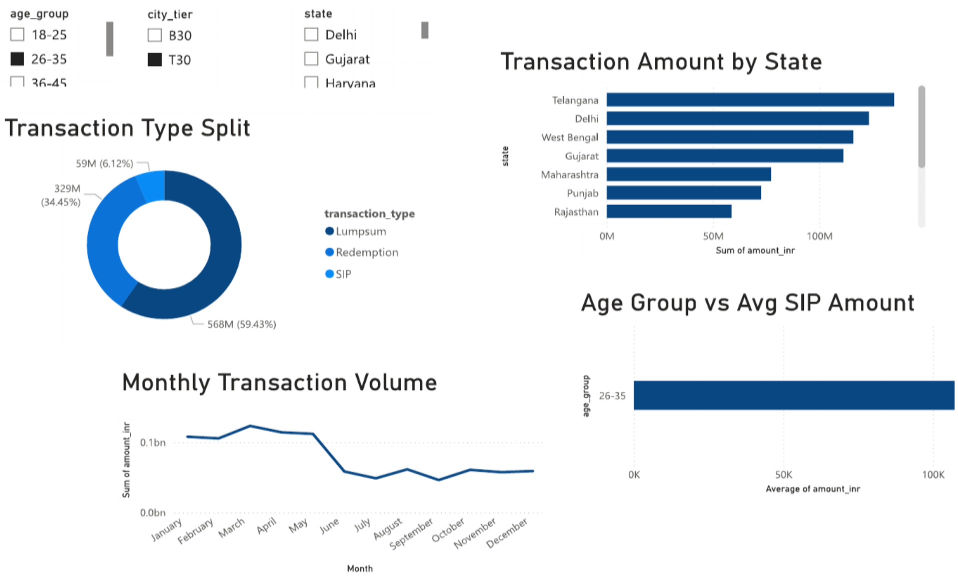
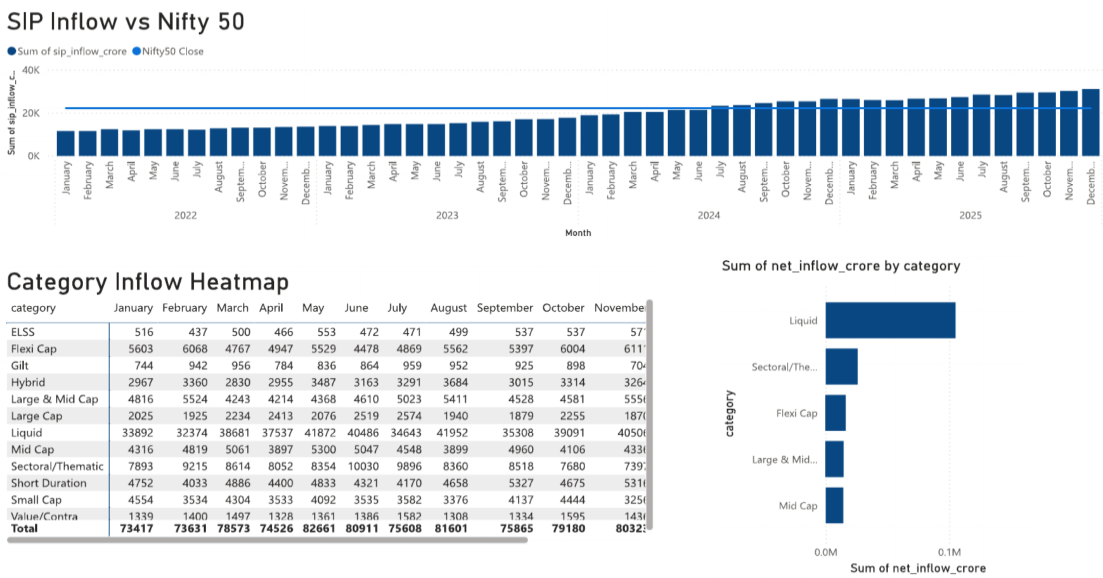


Figure 3: Investor Analytics — Transaction Type Split, State Distribution, Age Group Analysis

The Investor Analytics page shows a Transaction Type Split donut chart (SIP: 59.43%, Lumpsum: 34.45%, Redemption: 6.12%), a Transaction Amount by State horizontal bar chart (Telangana, Delhi, and West Bengal leading), a Monthly Transaction Volume line chart showing a slight decline from mid-year, and an Age Group vs Avg SIP Amount bar chart highlighting the 26–35 cohort.

Page 4 — SIP & Market Trends



category	January	February	March	April	May	June	July	August	September	October	Novembe
ELSS	516	437	500	466	553	472	471	499	537	537	57
Flexi Cap	5603	6068	4767	4947	5529	4478	4869	5562	5397	6004	611
Gilt	744	942	956	784	836	864	959	952	925	898	70
Hybrid	2967	3360	2830	2955	3487	3163	3291	3684	3015	3314	326
Large & Mid Cap	4816	5524	4243	4214	4368	4610	5023	5411	4528	4581	555
Large Cap	2025	1925	2234	2413	2076	2519	2574	1940	1879	2255	187
Liquid	33892	32374	38681	37537	41872	40486	34643	41952	35308	39091	4050
Mid Cap	4316	4819	5061	3897	5300	5047	4548	3899	4960	4106	433
Sectoral/Thematic	7893	9215	8614	8052	8354	10030	9896	8360	8518	7680	739
Short Duration	4752	4033	4886	4400	4833	4321	4170	4658	5327	4675	531
Small Cap	4554	3534	4304	3533	4092	3535	3582	3376	4137	4444	325
Value/Contra	1339	1400	1497	1328	1361	1386	1582	1308	1334	1595	143
Total	**73417**	**73631**	**78573**	**74526**	**82661**	**80911**	**75608**	**81601**	**75865**	**79180**	**8032**

Figure 4: SIP & Market Trends — SIP vs Nifty 50, Category Inflow Heatmap

The SIP & Market Trends page contains a SIP Inflow vs Nifty 50 combined chart (bar for SIP inflows, line for Nifty 50 close) from 2022–2025, a Category Inflow Heatmap table showing monthly inflows across all fund categories, and a Net Inflow by Category bar chart on the right highlighting Liquid as the dominant category for net inflows.

Key Findings

The following findings were derived from the combined EDA, Performance Analytics, and Advanced Analytics work completed in this project:

Industry & Market Findings

- The Indian mutual fund industry showed strong and consistent growth from 2022 to 2025, with total AUM nearly doubling from approximately 30 lakh crore to 62.74 lakh crore.
- SIP inflows grew steadily year-over-year, reflecting increasing retail investor participation and systematic investment culture.
- SBI Mutual Fund holds the largest market share by AUM among the 10 fund houses analyzed, followed by ICICI Prudential MF and HDFC Mutual Fund.
- Industry folio count grew strongly between 2022 and 2025, indicating new investor onboarding.

Fund Performance Findings

- Funds AMFI 120505, 100033, 119094, 119598, and 148567 ranked as top-5 composite performers based on the weighted scorecard of CAGR, Sharpe Ratio, Alpha, Expense Ratio, and Maximum Drawdown.
- AMFI 148567 achieved the highest Sharpe Ratio (1.45) among all analyzed funds, indicating the best risk-adjusted return relative to the 6.5% risk-free rate.
- The majority of fund returns are positively correlated with each other, limiting the diversification benefit of holding multiple equity schemes.
- Tracking errors for active equity funds (0.19–0.28 annualized) confirm meaningful deviation from the NIFTY100 benchmark.

Investor Behavior Findings

- SIP is the dominant transaction type, accounting for 59.43% of all transactions by value.
- The 26–35 age group has the highest average SIP amount, making them the most valuable investor segment.
- Telangana, Delhi, and West Bengal are the top three states by total investment amount.
- T30 (Top 30) cities contribute a disproportionately higher share of total AUM compared to B30 cities.
- Investor cohort analysis identified 2024 entrants as the highest-volume cohort.

Risk & Portfolio Findings

- VaR and CVaR analysis identified funds with concentrated tail risk, enabling differentiation between risk profiles.
- HHI-based sector concentration analysis flagged several funds with high exposure to Financial Services and Technology sectors.